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Sample-efficient quantum error mitigation via classical learning surrogates

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The pursuit of practical quantum utility on near-term quantum processors is critically challenged by their inherent noise. Quantum error mitigation (QEM) techniques are leading solutions to improve computation fidelity with relatively low qubit-overhead, while full-scale quantum error correction remains a distant goal. However, QEM techniques incur substantial measurement overheads, especially when applied to families of quantum circuits parameterized by classical inputs. Focusing on zero-noise extrapolation (ZNE), a widely adopted QEM technique, here we devise the surrogate-enabled ZNE (S-ZNE), which leverages classical learning surrogates to perform ZNE entirely on the classical side. Unlike conventional ZNE, whose measurement cost scales linearly with the number of circuits, S-ZNE requires only constant measurement overhead for an entire family of quantum circuits, offering superior scalability. Theoretical analysis indicates that S-ZNE achieves accuracy comparable to conventional ZNE in many practical scenarios, and numerical experiments on up to 100-qubit ground-state energy and quantum metrology tasks confirm its effectiveness. Our approach provides a template that can be effectively extended to other quantum error mitigation protocols, opening a promising path toward scalable error mitigation.

INTRODUCTION

Recent advances in quantum hardware fabrication have enabled demonstrations of computational capabilities approaching or even surpassing the classical frontier [1–6]. However, the utility of these modern quantum processors is intrinsically limited by decoherence and operational errors, which degrade computational fidelity and obscure potential quantum utility [7–9]. While full quantum error correction remains infeasible in the foreseeable future due to its immense resource overhead [10], quantum error mitigation (QEM) has emerged as the critical strategy to suppress the noise-induced bias in expectation values on noisy hardware [11–14]. On the one side, recent progress has validated QEM’s indispensable role in many realistic applications, including ground-state energy estimation [15, 16] and digital quantum simulation [17–19]. On the other side, its principles are also poised to reduce the resource overhead of implementing quantum error correction protocols [20–30].

QEM has witnessed rapid theoretical and experimental advancements in recent years, yielding a diverse spectrum of strategies. Notable among them are zero-noise extrapolation (ZNE) [18, 31–36], probabilistic error cancellation [37–40], virtual distillation [41–44], symmetry constraints [45, 46], learning-based approaches [47–51], and quantum subspace expansion [52, 53], among others. Owing to its conceptual simplicity and model-agnostic nature, ZNE has been successfully deployed on large-scale processors [18], such as 127-qubit systems, to explore the computational utility of near-term quantum devices. Nevertheless, the practical adoption of QEM faces significant challenges, the most critical being the

formidable sampling overhead required for its implementation [54, 55]. This sampling issue becomes particularly prohibitive for practical quantum tasks that involve families of circuits parameterized by classical inputs, as seen in variational quantum algorithms [56–58], quantum many-body simulations [59–61], and quantum sensing [62–65] applications. In such scenarios, each distinct parameter value necessitates an independent and often costly execution of the full QEM protocol, leading to a potentially unsustainable resource burden. Consequently, a pivotal direction for current research is the development of efficient QEM frameworks that can mitigate errors across an entire family of quantum circuits with reduced sampling costs.

To address this bottleneck, we propose the surrogate-enabled ZNE (S-ZNE) framework. The core of S-ZNE is leveraging classical learning surrogates [66–68] to decouple data acquisition from quantum execution. While it follows the standard ZNE procedure that extrapolates from amplified noise levels, the key difference is that the requisite noisy expectation values are predicted by the surrogate rather than measured directly. Consequently, after an initial training phase of the surrogates, S-ZNE mitigates errors for a class of parametrized circuits with different inputs purely on the classical side, without any additional measurement overhead. This constant-overhead property provides a fundamental scaling advantage over conventional ZNE, whose resource demands grow linearly with respect to the number of tasks.

The advantages of S-ZNE are established through both rigorous theoretical analysis and extensive numerical validation. Theoretically, we prove that under many practical scenarios, S-ZNE achieves an error scaling that is provably comparable to conventional ZNE. In addition, we conduct systematic experiments to validate S-ZNE’s performance on two distinct applications: ground state

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energy estimation and quantum metrology task [69] on up to 100 qubits. The achieved results indicate that S-ZNE achieves performance comparable to that of conventional ZNE while significantly reducing the sampling overhead. These results position S-ZNE as a powerful and practical tool for enhancing the capabilities of near-term quantum computing.

RESULTS

Problem setup. A crucial subroutine in a wide range of quantum algorithms involves the mean-value estimation for a family of parametrized quantum states controlled by classical inputs, i.e., $\{\rho(\mathbf{x})|\mathbf{x} \in [0, 2\pi]^d\}$. Here $\rho(\mathbf{x})$ is generated by applying a parametrized quantum circuit $U(\mathbf{x})$ to a fixed initial state ρ_0 with $\rho(\mathbf{x}) = U(\mathbf{x})\rho_0 U(\mathbf{x})^\dagger$. Concrete examples in this context include digital quantum simulation [70], variational quantum algorithms [57], and quantum metrology [69]. Without loss of generality, $U(\mathbf{x})$ can always be decomposed into a set of Clifford gates (i.e., $\{H, S, \text{CNOT}\}$) and rotational gates along the Z-axis (i.e., $\text{RZ}(\mathbf{x})$), which form a universal gate set [71]. Mathematically, we have

$$U(\mathbf{x}) = \left(\prod_{j=1}^d C_j \text{RZ}(\mathbf{x}_j) \right) C_0, \quad (1)$$

where C_j refers to the unitary at the j -th layer formed by Clifford gates, and d denotes the dimension of classical inputs $\mathbf{x}$. The mean-value outcome of $\rho(\mathbf{x})$ for a given observable O is denoted by $f(\mathbf{x}, O) = \text{Tr}(\rho(\mathbf{x})O)$. When deployed on noisy quantum processors, such parametrized states are often corrupted by a noise channel $\mathcal{N}_\lambda(\cdot)$ with λ being the noise level. Accordingly, the experimentally accessible value is

$$f(\mathbf{x}, O, \lambda) = \text{Tr}(\mathcal{N}_\lambda(\rho(\mathbf{x}))O) \quad (2)$$

instead of $f(\mathbf{x}, O)$. Throughout this study, we model the dominant noise as Pauli channels, as any decoherent noise process can be transformed into Pauli noise through Pauli twirling [11].

Zero-noise extrapolation (ZNE) is a technique designed to estimate the noiseless mean-value outcome $f(\mathbf{x}, O)$ by strategically leveraging the noisy estimations in Eq. (2). As shown in Fig. 1, the ZNE protocol involves systematically amplifying the noise channel $\mathcal{N}_\lambda$ with the level λ , e.g., via unitary folding [72]. Denote $\mathcal{N}_0$ the identity channel, which leaves any quantum state unchanged. The ideal expectation value yields $f(\mathbf{x}, O) \equiv f(\mathbf{x}, O, \lambda = 0)$. In practice, the noisy level λ can only be amplified rather than reduced. Accordingly, ZNE obtains expectation values at a series of elevated effective noise levels $\{\lambda_j\}_{j=1}^u \in \mathbb{N}_+$ with $\lambda_j < \lambda_{j+1}$. After that, the desired noiseless result $f(\mathbf{x}, O)$ is estimated through extrapolation. Specifically, let $g(\cdot)$ be the employed extrapolation function and $\hat{f}(\mathbf{x}, O, \lambda_j)$ be an estimation of each noisy expectation $f(\mathbf{x}, O, \lambda_j)$ with M snapshots. The estimation of the zero-noise limit, i.e., $f(\mathbf{x}, O)$, yields

$$g(\mathbf{z}_C(\mathbf{x})) \equiv g\left(\left[\hat{f}(\mathbf{x}, O, \lambda_1), \dots, \hat{f}(\mathbf{x}, O, \lambda_u)\right]\right). \quad (3)$$

The entire ZNE procedure is inherently measurement-intensive. That is, as each artificially amplified noise level λ_j must be independently sampled, the total sampling cost grows linearly with the boost factor u . In addition, when applying ZNE to mitigate $f(\mathbf{x}, O)$, the overhead further scales with the number of classical inputs $\mathbf{x}$, rendering a substantial sampling overhead.

Implementation of S-ZNE. To alleviate the substantial measurement overhead of ZNE when estimating $f(\mathbf{x}, O)$ over varying inputs $\mathbf{x}$, we propose the surrogate-ZNE (S-ZNE), a framework that leverages classical learning surrogates to predict $f(\mathbf{x}, O, \lambda_j)$ and thereby circumvent repeated quantum hardware evaluations. For clarity, let us first recap the mechanism of classical learning surrogates, followed by presenting the implementation of S-ZNE.

The objective of classical learning surrogates [68] is to effectively predict the expectation values of target observables across $\{\rho(\mathbf{x})\}$. An attractive feature of this approach is that once the dataset is collected from quantum hardware, both the training and inference processes are entirely classical. For example, the training dataset $\mathcal{T} = \{\mathbf{x}^{(i)}, y^{(i)}\}_{i=1}^n$ can be constructed by directly measuring the expectation values on quantum hardware, where $\mathbf{x}^{(i)}$ refers to the i -th training example and $y^{(i)}$ is the estimated expectation value of the observable O for the (noisy) state $\rho(\mathbf{x}^{(i)})$ with T measurement shots. Once $\mathcal{T}$ is constructed, the regression-based classical learning surrogate can be effectively implemented, i.e.,

$$h(\mathbf{x}, O) = \langle \Phi_{\mathcal{C}(\Lambda)}(\mathbf{x}), \mathbf{w} \rangle, \quad (4)$$

where $\Phi_{\mathcal{C}(\Lambda)}(\mathbf{x})$ denotes the feature vector composed of trigonometric basis functions with frequencies truncated at Λ . The weight vector $\mathbf{w}$ is obtained by solving the ridge regression problem with a regularization hyperparameter $\gamma > 0$, i.e.,

$$\min_{\mathbf{w}} \left\{ \frac{1}{n} \sum_{i=1}^n \left(y^{(i)} - \langle \Phi_{\mathcal{C}(\Lambda)}(\mathbf{x}^{(i)}), \mathbf{w} \rangle \right)^2 + \gamma \|\mathbf{w}\|_2^2 \right\}.$$

Refer to Method and SI A for implementation details.

An overview of S-ZNE is illustrated in Fig. 1. The philosophy of S-ZNE is to decouple noise extrapolation from repeated quantum executions by using classical surrogate models to predict circuit outcomes for new inputs. Mathematically, denote the learning surrogate trained on $f(\mathbf{x}, O, \lambda_j)$ as $h(\mathbf{x}, O, \lambda_j)$. For any new input $\mathbf{x}'$, the surrogate outputs $h(\mathbf{x}', O, \lambda_j)$ as the prediction of $f(\mathbf{x}', O, \lambda_j)$ instead of $\hat{f}(\mathbf{x}', O, \lambda_j)$ in Eq. (3). As the prediction is conducted purely on the classical side, S-ZNE is able to substantially reduce the sampling overhead.

The implementation of S-ZNE comprises three steps, which are data collection, surrogate modeling, and ZNE estimation. At the first stage, S-ZNE performs a one-time investment in quantum measurements to construct the training $\mathcal{T}(\lambda_j) = \{\mathbf{x}^{(i,j)}, y^{(i,j)}\}_{i=1}^{n_j}$ at each noise level λ_j , where n_j denotes the number of training examples used in $h(\mathbf{x}, O, \lambda_j)$ and $y^{(i,j)}$ refers to the mean-value es-

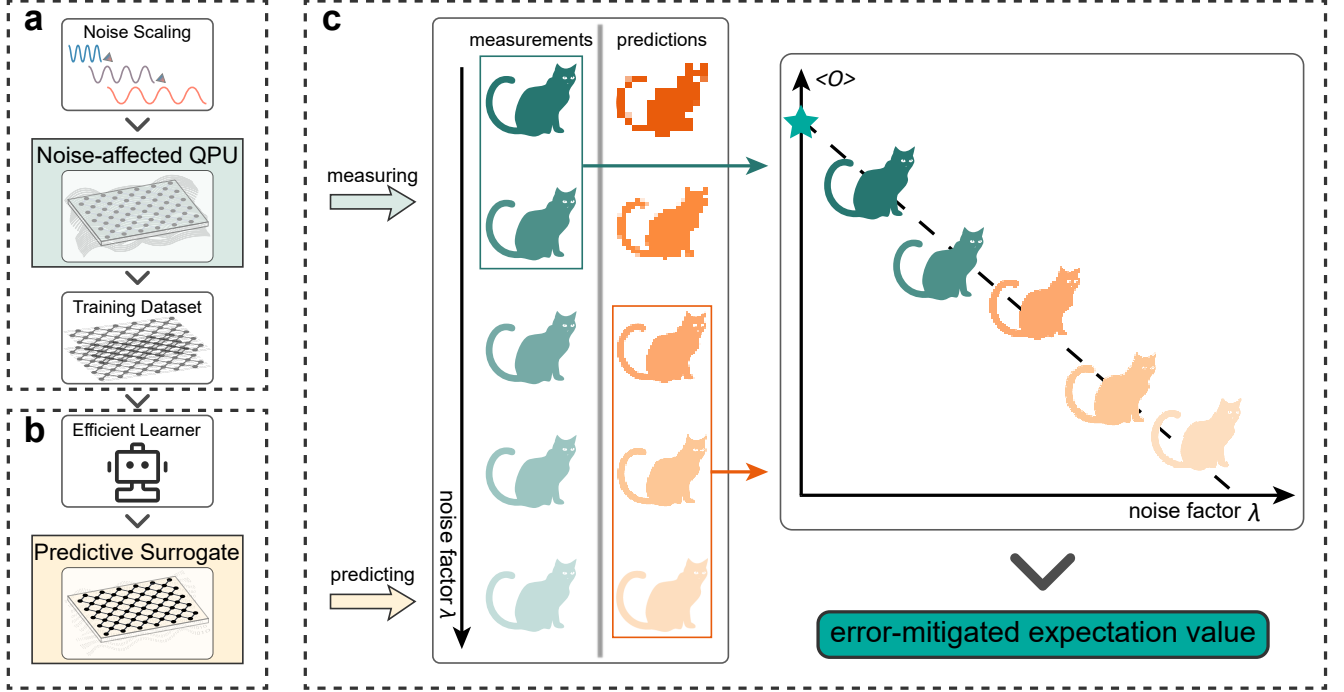


FIG. 1. **The scheme of surrogate-enabled zero noise extrapolation (S-ZNE).** The S-ZNE framework towards a class of parametrized quantum circuits comprises three key stages. **a.** Data Collection. Noise scaling (e.g., via unitary folding) is applied to generate the training dataset. At each noise level, different classical inputs are fed into the quantum circuit, followed by quantum sampling. **b.** Surrogate Modeling. The constructed dataset trains classical learning surrogates to estimate expectation values under target noise conditions purely on the classical side. **c.** Surrogate-Enabled Extrapolation: Given any new input, the optimized classical learning surrogates at all noise factors λ can be used to predict the corresponding expectation values. In the main text, we use surrogate prediction for all noise levels, while a hybrid approach combining surrogate predictions and direct measurements from noisy quantum processor (hybrid S-ZNE) is discussed in SI E. These results extrapolate to $\lambda \rightarrow 0$, yielding an error-mitigated estimate with reduced quantum overhead.

timination of $f(\mathbf{x}, O, \lambda_j)$ with T snapshots. Once u training datasets are prepared, S-ZNE separately implement u classical learning surrogates $\{h(\mathbf{x}, O, \lambda_j)\}_{j=1}^u$ following Eq. (4). Last, S-ZNE employs the optimized u classical learning surrogates to complete ZNE estimation purely on the classical side. According to Eq. (3), the zero-noise limit $f(\mathbf{x}, O)$ for any new input $\mathbf{x}$ is estimated by

$$g(\mathbf{z}_S(\mathbf{x})) = g([h(\mathbf{x}, O, \lambda_1), \dots, h(\mathbf{x}, O, \lambda_u)]).$$

Refer to Methods for more elaborations.

We analyze the estimation error of S-ZNE relative to the conventional ZNE, as stated in the theorem below, whose proof is deferred to SI B.

Theorem 1 (Informal). *Following notations in Eqs. (1)-(4), let L be the Lipschitz constant of $g(\cdot)$, and ζ^2 be the intrinsic extrapolation error. Suppose that $\mathbf{x} \in [-R, R]^d$ is sampled from a distribution $\mathbb{D}$. When the conventional ZNE costs M measurements to obtain $\hat{f}(\mathbf{x}, O, \lambda_j)$, with probability at least $1 - 0.05u$, its average performance yields $\mathbb{E}_{\mathbf{x}}|f(\mathbf{x}, O) - g(\mathbf{z}_C(\mathbf{x}))|^2 \leq \zeta^2 + \mathcal{O}(L^2 u B^2 / M)$. In addition, when $\Lambda > \text{deg}(1 + R)$ and $n_j \geq \tilde{O}(B^2 M (\text{deg}/\Lambda)^{4\Lambda})$, with probability at least $1 - u\delta$,*

the average performance of S-ZNE is upper bounded by $\mathbb{E}_{\mathbf{x}}|f(\mathbf{x}, O) - g(\mathbf{z}_S(\mathbf{x}))|^2 \leq \zeta^2 + \mathcal{O}(L^2 u B^2 / M)$.

The achieved results deliver several implications. First, the performance of both conventional ZNEs and S-ZNE depends on ζ^2 . This error term stems from the choice of the extrapolation function $g(\cdot)$ and is independent of the approach used to estimate expectation values $f(\mathbf{x}, O, \lambda_j)$ for the varied j . Moreover, the performance of both protocols is influenced by the Lipschitz constant L . As derived in SI B, for the linear extrapolation, L does not dominate the overall error, which scales approximately as $L \rightarrow 4/u$ as u increases. Besides, the derived sample complexity n_j reflects the efficiency of S-ZNE. In many realistic settings, the truncation parameter Λ remains a small constant, e.g., noisy circuits with a large q and quantum simulation tasks characterized by a small R . Under such conditions, S-ZNE is computationally efficient and can achieve performance comparable to that of conventional ZNE.

Remark. The proposed S-ZNE framework is highly flexible and can be adapted to a wide range of quantum tasks. For example, the regression-based surrogate can be substituted with other provably efficient learning models. In

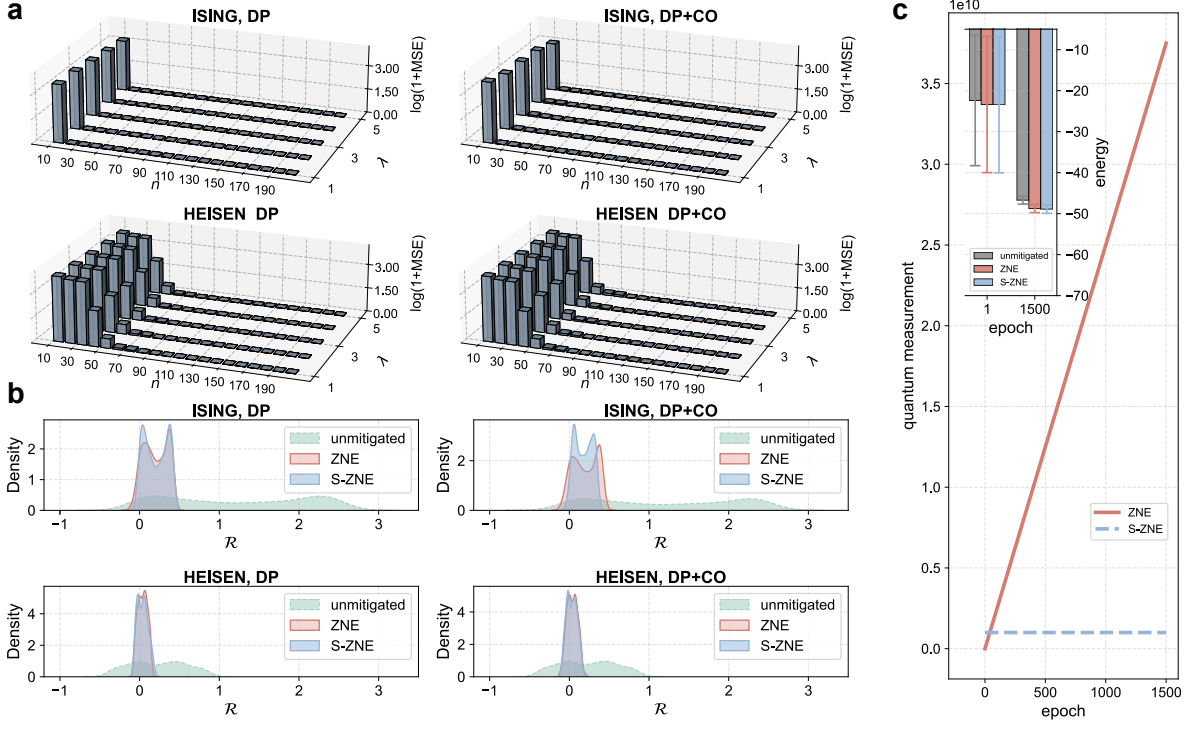


FIG. 2. **Performance of S-ZNE in ground state energy estimation for quantum many-body systems.** **a.** Surrogate model prediction accuracy under varying training set sizes (n_j) and noise levels (λ_j). Prediction error, quantified by $\log(1 + \text{MSE})$, is evaluated on 1000 test samples for both transverse field Ising model (TFIM) and Heisenberg model (HM) under depolarizing (DP) and combined depolarizing+coherent (DP+CO) noise channels. **b.** Distribution of residuals (R) of unmitigated, S-ZNE, and conventional ZNE protocols, evaluated by the kernel density estimation over 1000 test configurations. **c.** Sampling overhead analysis during the optimization. The main panel shows cumulative measurement costs for S-ZNE (constant overhead) versus ZNE (linear scaling with iteration). Inset displays estimated ground-state energies at epochs 1 and 1500 for TFIM (exact value: -50.50), demonstrating comparable convergence between S-ZNE and ZNE despite significantly reduced quantum resource requirements for S-ZNE.

addition, S-ZNE can be extended to a hybrid structure, where classical learning surrogates are employed only at high noise levels to further balance accuracy and efficiency. Refer to SI C for more details.

Numerical Results. Our first task is exploring the capabilities of S-ZNE in ground-state energy estimation using variational quantum algorithms (VQAs) [57]. Given a specified ansatz $U(\mathbf{x})$, VQA estimates the ground-state energy of the Hamiltonian H by iteratively updating $\mathbf{x}$ through gradient descent to minimize $f(\rho(\mathbf{x}), H)$. Due to system noise, each iteration requires QEM to obtain reliable gradient information, incurring an increased sampling burden. To this end, S-ZNE offers an effective alternative to overcome this limitation.

To comprehend the effectiveness of S-ZNE, we consider two standard quantum many-body systems, i.e., the one-dimensional transverse field Ising models (TFIMs) and Heisenberg model (HMs) with open boundary conditions and $N = 100$ qubits. Their mathematical forms are given by $H_{\text{Ising}} = -\sum_i J Z_i Z_{i+1} - \sum_i h X_i$ and $H_{\text{Heisen}} = \sum_i (J_x X_i X_{i+1} + J_y Y_i Y_{i+1} + J_z Z_i Z_{i+1})$, respectively. For TFIM, we set $J = 0.1$ and $h = 0.5$, while for HM, we set $J_x = 0.1$, $J_y = 0.5$ and $J_z = 0$. For VQAs, we adopt

the Hamiltonian variational ansatz [73], where the implementation of $U(\mathbf{x})$ depends on the explored Hamiltonian. The system noise is modeled by both the global depolarizing (DP) channel and the coherent (CO) channel. The depolarization rate is set as $1 - (1 - p_g)^{\lambda_j}$ with $p_g = 0.05$. The coherent error is modeled by introducing perturbations on the input $\mathbf{x}$, with the perturbation on each entry drawn uniformly from the range $[-0.01 \cdot \lambda_j, 0.01 \cdot \lambda_j]$. The extrapolation function $g(\cdot)$ is realized through a linear least-squares fit. Refer to SI E for more details and additional results under diverse settings.

We begin by assessing the predictive accuracy of classical surrogates $\{h(\mathbf{x}, H, \lambda_j)\}$ with $H \in \{H_{\text{Ising}}, H_{\text{Heisen}}\}$ with varying the number training examples n_j . The prediction performance is quantified by $\log(1 + \text{MSE})$ over a test dataset with 1000 unseen inputs, where MSE refers to the mean-square error. As shown in Fig. 2a, for both DP and DP+CO noisy settings, the prediction error rapidly decreases with the increased n_j . When n_j exceeds 100, the prediction error for both Hamiltonians is ignorable. In addition, the prediction error exhibits only a weak dependence on the noise level λ_j , slightly decreasing at higher noise levels.

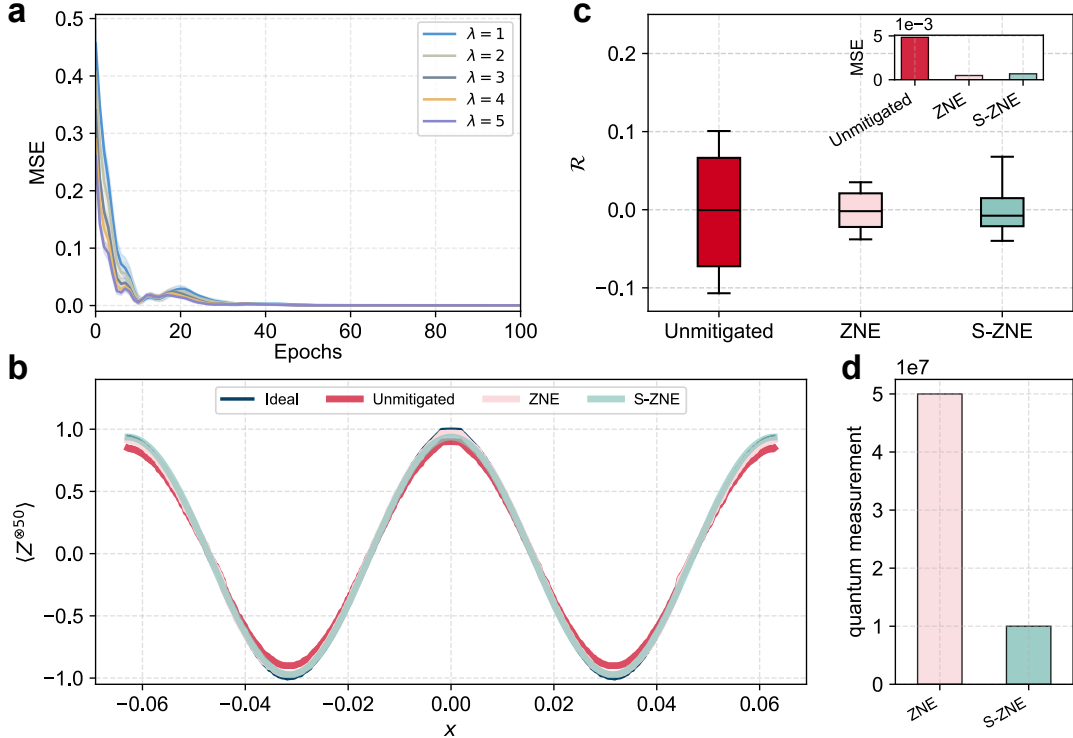


FIG. 3. **Application of S-ZNE to GHZ-state quantum metrology.** **a.** Training dynamics of learning surrogates under varied noise levels λ_i . **b.** Reconstructed phase estimation signals after quantum error mitigation. S-ZNE predictions and ZNE estimates both recover the ideal $2\pi/100$ -periodic response. **c.** Residual analysis of error-mitigated expectation values. Both S-ZNE and ZNE maintain residuals close to the ideal $\cos(100x)$ signal, with an inset comparing overall MSE performance. **d.** Sampling overhead of ZNE and S-ZNE. The y-axis refers to the cumulative number of measurements consumed by S-ZNE/ZNE for error mitigation.

We next compare the error mitigation performance between S-ZNE and ZNE by evaluating their residuals, i.e., $\mathcal{R}_S = g(\mathbf{z}_S(\mathbf{x})) - f(\mathbf{x}, \mathbf{H})$ and $\mathcal{R}_C = g(\mathbf{z}_C(\mathbf{x})) - f(\mathbf{x}, \mathbf{H})$, respectively. For all surrogates used in S-ZNE, the number of training examples is fixed to be $n_j = 200$. For conventional ZNE, $M = 1 \times 10^6$ measurements are used to estimate each $\hat{f}(\mathbf{x}, \mathbf{H}, \lambda_j)$. The residual distributions, estimated by kernel density estimation (KDE) [74] over 1000 test samples, are adopted as the metric. As shown in Fig. 2b, S-ZNE residuals cluster tightly around zero, achieving a distribution statistically comparable to that of conventional ZNE. This demonstrates that S-ZNE successfully mitigates errors to a similar level as ZNE under both DP and DP+CO noisy settings.

We further analyze the consumed quantum resources for S-ZNE and ZNE. As shown in Fig. 2c, the inset shows a bar plot comparing the estimated ground-state energies of the TFIM at the 1st and 1500-th iterations for S-ZNE, conventional ZNE, and the unmitigated cases, averaged over 10 independent variational optimization runs. With the exact ground-state energy being -50.50 , the results indicate that both S-ZNE and conventional ZNE provide comparable and significantly improved accuracy compared to the unmitigated case. In terms of resource efficiency, S-ZNE requires only a constant sampling overhead, consuming 1×10^9 measurements throughout the

entire optimization process. In contrast, the conventional ZNE necessitates $M \times u$ quantum measurements for every evaluation of $g(\mathbf{z}_C(\mathbf{x}))$, resulting in a linearly increasing sampling overhead that becomes dramatically larger than that of S-ZNE. The total cost of ZNE accumulates to 3.75×10^{10} measurements over the optimization course, which is 37.5 times that of S-ZNE.

Our second task applies S-ZNE to quantum metrology based on Ramsey interferometry [69, 75]. The protocol uses a 100-qubit GHZ probe state,

$$|\text{GHZ}\rangle_{100} = \frac{1}{\sqrt{2}}(|0\rangle^{\otimes 100} + |1\rangle^{\otimes 100}).$$

The unknown phase x is encoded via the global unitary $U(x) = \text{RZ}(x)^{\otimes 100}$, which imprints a relative phase $\exp(i100x)$ between the two components of the GHZ state. The ideal signal is then given by

$$f(x, Z^{\otimes 100}) = \text{Tr}(\rho(x) Z^{\otimes 100}) = \cos(100x),$$

where $\rho(x) = U(x) |\text{GHZ}\rangle_{100} \langle \text{GHZ}| U^\dagger(x)$. See SI E for more details. In the subsequent experiments, the global depolarizing noise with the strength $p_g = 0.1$ is considered to damp the signal amplitude.

We compare S-ZNE with conventional ZNE using noisy data collected at $u = 5$ noise amplification levels. For

conventional ZNE, it estimates the noisy expectation values $\hat{f}(x, Z^{\otimes N}, \lambda_j)$ by using $M = 2 \times 10^4$ measurement shots per evaluation point. For S-ZNE, we train the learning surrogate in Eq. (4) with $n_j = 100$ training samples ($T = 2 \times 10^4$) for $\forall j \in [5]$. As shown in Fig. 3a, the MSE of the surrogates consistently drops below 2×10^{-4} within 100 iterations across all noise factors, despite widely varying initial errors. This rapid and uniform convergence confirms that the surrogates reliably learn the noise-parametrized response even in this high-dimensional setting.

With stable surrogates in place, we assess phase estimation accuracy in terms of the residual $\mathcal{R}$. Fig. 3b displays the error-mitigated expectation values over a representative range of x , which is discretized into 500 points for evaluation. The S-ZNE predictions, obtained by evaluating the trained surrogates at the inference stage, closely follow the ideal $\cos(100x)$ curve. Conventional ZNE, based on repeated quantum measurements, also recovers the signal accurately. Both methods preserve the $2\pi/100$ periodicity essential for Heisenberg-limited metrology. Quantitatively, as shown in Fig. 3c, conventional ZNE achieves an MSE of 4.83×10^{-4} , while S-ZNE attains an MSE of 6.79×10^{-4} —both significantly lower than the unmitigated MSE of 4.84×10^{-3} .

Critically, after initial training, S-ZNE mitigates errors classically with constant overhead, fundamentally outperforming conventional ZNE’s linear scaling with input number. In practical terms, conventional ZNE required $500 \times u \times M = 5 \times 10^7$ total measurement shots to evaluate all 500 phase values across 5 noise levels, while S-ZNE needed only $n_j \times u \times T = 1 \times 10^7$ shots during training—yielding a 80% reduction in quantum measurement cost. Crucially, this efficiency gap widens as the number of input points increases.

DISCUSSION

In this study, we introduced S-ZNE, a framework that substantially reduces the measurement overhead associated with conventional ZNE when applied to families of quantum circuits parameterized by classical inputs. Experimental results on ground-state energy estimation and quantum metrology tasks indicate that S-ZNE attains comparable extrapolation accuracy with conventional ZNEs while significantly reducing the measurement cost. These findings establish S-ZNE as a practical tool for enhancing QEM capabilities on near-term quantum processors.

Several promising research avenues emerge from this work, spanning both QEM and classical learning surrogates. An important direction is extending classical learning surrogates to other QEM strategies beyond ZNE, such as virtual distillation [41, 44], probabilistic error cancellation [37], and learning-based approaches [47]. Furthermore, integrating learning surrogates with experimental calibration procedures—such as real-time noise regime adaptation or online calibration—could enhance the model’s practicality and robustness in experimental settings. Exploring the synergy between deep-learning-

based surrogates [76–80] and these adaptive protocols may also open new avenues for developing more responsive and theoretically-grounded quantum error mitigation techniques.

METHODS

Noise models and amplification. In numerical simulations, we employ two primary noise models: global depolarizing noise and coherent noise. The global depolarizing channel acts collectively on all N qubits and represents a symmetric and maximally mixing noise process. The channel is defined by $\mathcal{N}_g(\rho) = (1-p_g)\rho + p_g \frac{\mathbb{I}^{\otimes N}}{2^N}$, where $p_g \in [0, 1]$ is the global depolarizing rate and $\mathbb{I}^{\otimes N}$ denotes the N -qubit identity operator. This noise model preserves the identity component while uniformly damping all other elements of the density matrix.

The coherent error arises from systematic miscalibrations in quantum gate operations. We model these errors as small, stochastic perturbations of rotation angles in single-qubit rotational gates. More precisely, given a rotation gate $R_P(x) = e^{-ixP/2}$ with $P \in \{X, Y, Z\}$, its noisy implementation becomes $R_P(x) \mapsto e^{-i(x+\theta_P)P/2}$, where the angular offset θ_P for each gate instance is drawn independently from a uniform distribution over $[-\epsilon, \epsilon]$. Refer to SI A for more details.

To enable noise extrapolation, we amplify both noise types by increasing their effective strength. For global depolarizing noise, we apply unitary folding, which effectively applies the base depolarizing channel λ_j times, yielding an amplified depolarizing rate $p_{\lambda_j} = 1 - (1 - p_g)^{\lambda_j}$. For coherent noise, we directly increase the noise strength by scaling the perturbation range to $[-\lambda_j\epsilon, \lambda_j\epsilon]$.

Implementation of Learning Surrogates. The S-ZNE framework employs classical learning surrogates to predict noisy expectation values $f(\mathbf{x}, O, \lambda_j)$ at amplified noise levels λ_j . Our primary implementation adopts ridge regression due to its computational efficiency and theoretical guarantees. We also explore kernel-based surrogates with classical shadows as a comparative approach (see SI E). Below, we outline both methods and provide full details in SI A.

Ridge-Based Regression Surrogate. For quantum circuits with *correlated parameters*, we use ridge regression on an explicit trigonometric feature map. Here, the d -dimensional parameter vector $\mathbf{x}$ is partitioned into S groups $\{\mathbf{r}_1, \dots, \mathbf{r}_s\}$, where parameters in each group are identical.

Given a training set $\mathcal{T}(\lambda_j) = \{\mathbf{x}^{(i,j)}, y^{(i,j)}\}_{i=1}^{n_j}$ at noise level λ_j , the surrogate model is:

$$h(\mathbf{x}, O, \lambda_j) = \langle \Phi_{\mathcal{C}(\Lambda)}(\mathbf{x}), \mathbf{w}_j \rangle, \quad (5)$$

where $\Phi_{\mathcal{C}(\Lambda)}(\mathbf{x})$ is a feature vector composed of trigonometric monomials defined via a frequency-truncated set $\mathcal{C}(\Lambda) = \{\boldsymbol{\omega} \in \{0, \pm 1\}^d \mid \|\boldsymbol{\omega}\|_0 \leq \Lambda\}$.

To capture parameter correlations, the feature map is constructed group-wise. Specifically, for a frequency vector $\boldsymbol{\omega}$, whose components $\omega_{s,k}$ correspond to the k -th

parameter in group s , the feature map is defined as:

$$\Phi_{\omega}(\mathbf{x}) = \prod_{s=1}^S \left[\cos(\mathbf{r}_s)^{N_s^+(\omega)} \cdot \sin(\mathbf{r}_s)^{N_s^-(\omega)} \right], \quad (6)$$

where $N_s^+(\omega) = \sum_{k=1}^{d_s} \mathbb{1}\{\omega_{s,k} = 1\}$ and $N_s^-(\omega) = \sum_{k=1}^{d_s} \mathbb{1}\{\omega_{s,k} = -1\}$ count the occurrences of cosine and sine terms in group s , respectively, with d_s being the number of parameters in group s .

The weight vector $\mathbf{w}_j$ is obtained by solving the ridge regression problem:

$$\min_{\mathbf{w}_j} \left\{ \frac{1}{n_j} \sum_{i=1}^{n_j} \left(y^{(i,j)} - \langle \Phi_{\mathbf{c}(\Lambda)}(\mathbf{x}^{(i,j)}), \mathbf{w}_j \rangle \right)^2 + \gamma \|\mathbf{w}_j\|_2^2 \right\}, \quad (7)$$

with regularization hyperparameter $\gamma > 0$.

Kernel-Based Learning Surrogate. For quantum circuits with *uncorrelated parameters*, S-ZNE can also be implemented using a kernel-based surrogate trained on classical shadows. At noise level λ_j , the training data $\mathcal{T}(\lambda_j) = \{\mathbf{x}^{(i,j)}, \tilde{\rho}_T(\mathbf{x}^{(i,j)})\}_{i=1}^{n_j}$ consists of n_j input-shadow pairs, where $\tilde{\rho}_T(\mathbf{x})$ is the shadow state reconstructed from T snapshots. The surrogate prediction is given by:

$$h(\mathbf{x}, O, \lambda_j) = \frac{1}{n_j} \sum_{i=1}^{n_j} \kappa_{\Lambda}(\mathbf{x}, \mathbf{x}^{(i,j)}) \cdot \text{Tr}(\tilde{\rho}_T(\mathbf{x}^{(i,j)})O), \quad (8)$$

using the truncated trigonometric kernel:

$$\kappa_{\Lambda}(\mathbf{x}, \mathbf{x}') = \sum_{\omega \in \mathbf{c}(\Lambda)} 2^{\|\omega\|_0} \Phi_{\omega}(\mathbf{x}) \Phi_{\omega}(\mathbf{x}'). \quad (9)$$

with feature map $\Phi_{\omega}(\mathbf{x})$ defined as:

$$\Phi_{\omega}(\mathbf{x}) = \prod_{l=1}^d \begin{cases} 1 & \text{if } \omega_l = 0, \\ \cos(x_l) & \text{if } \omega_l = 1, \\ \sin(x_l) & \text{if } \omega_l = -1. \end{cases} \quad (10)$$

Experimental and hyperparameter settings. All simulations shared the following common settings. All simulations are implemented using the **TensorCircuit** package [81] with its matrix product state (MPS) simulator backend, built on JAX (v0.4.30), for efficient simulation of large-qubit circuits. Quantum simulations used complex128 data types, while all classical learning surrogates at the training and prediction stages employed FP64 precision. Classical computations were performed on a server equipped with an NVIDIA A100 GPU (80GB HBM2) and dual Intel Xeon Gold 6348 CPUs, using CUDA 12.2 for acceleration.

In the task of ground state energy estimation, the base global depolarizing rate was set to $p_g = 0.05$. For simulations with combined depolarizing and coherent noise models, the coherent error perturbation range was set to $[-0.01, 0.01]$ at the base noise level, scaling linearly with the amplification factor λ . For regression-based surrogates, they were trained using the Adam optimizer with a learning rate of 0.01, employing automatic differentiation for gradient computation. The training used $T = 1 \times 10^6$ measurement shots per training sample over 1000 training epochs, with a regularization coefficient of $\gamma = 1$. The execution of VQAs in Fig. 2 employs the Adam optimizer with a learning rate of 0.001 over 1,500 training epochs, where gradients are approximated using finite-difference methods. Both the objective function $f(\mathbf{x}, H)$ and its gradient $\nabla f(\mathbf{x}, H)$ were estimated using ZNE extrapolation at each optimization step. The per-iteration measurement cost for conventional ZNE was $(2S+1) \times u \times M$, where $S = 2$ (parameters for finite-difference), $u = 5$ (noise levels), and $M = 1 \times 10^6$ (measurements per evaluation), resulting in a cumulative measurement cost of $1,500 \times (2 \times S + 1) \times u \times M = 3.75 \times 10^{10}$ shots over the entire optimization.

In the quantum metrology task, the base global depolarizing rate was set to $p_g = 0.1$. The regression-based learning surrogates were trained using the Adam optimizer with a learning rate of 0.01. The training used $T = 20,000$ measurement shots per training sample over 100 training epochs, with a regularization coefficient of $\gamma = 1$.

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DATA AVAILABILITY STATEMENT

The code and data are available from the corresponding author upon reasonable request.

AUTHOR CONTRIBUTION

H.-L. H. and Y. D. supervised the work. W.-Y. L., Y. D. G. Y. Y. S., and H.-L. H. proved the theoretical results. W.-Y. L., Y. D., and H.-L. H. performed the experiments and results analysis. All authors contributed to the writeup.

COMPETING INTERESTS

The authors declare no competing interests

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